Preterm with Respiratory Distress

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Introduction

Respiratory distress syndrome (RDS) and transient tachypnea of newborn (TTNB) are the most common causes of respiratory distress (RD) in preterm neonates. RDS is not just restricted to very and extremely preterm neonates, but also seen in late preterm and early term neonates. The various causes of respiratory distress and their associated signs and symptoms are listed in **Table 1**.

TABLE 1: Causes of RD in preterm neonates other than RDS and associated signs/symptoms.

Upper airway	***Symptoms/signs***	***Parenchymal***	***Symptoms/signs***	***Others***	***Symptoms/signs***
Subglottic stenosis	Severe suprasternal retractions. Stridor may or may not be present, difficult intubation and weak cry in severe cases	Congenital pneumonia	History of maternal risk factors for sepsis/ chorioamnionitis Onset of RD might be beyond 6 hours of life	*Cardiac:* Congenital heart diseases and PPHN (primary/ secondary)	Silent tachypnea and no improvement with 100% oxygen supplementation Preductal/post-ductal SpO_2 may or may not have a significant difference (>3%)
Congenital bilateral vocal cord palsy	History of perinatal asphyxia, antenatally diagnosed posterior fossa lesions, prolonged invasive mechanical ventilation	Air leak syndromes	History of receiving positive pressure ventilation, decreased air entry, and transillumination+	*Metabolic:* Hypothermia, hypoglycemia, metabolic acidosis, polycythemia, and hyperthermia	Anticipation based on risk factors, nonspecific symptoms
Subglottic hemangioma	Hemangiomas may be present in the skin	Congenital airway malformation and congenital diaphragmatic hernia	Might have been detected in antenatal scan or an incidental finding in radiography	*CNS causes:* Seizures, perinatal asphyxia, IVH, and maternal medications (anesthetics and $MgSO_4$)	Antenatal history is of vital importance

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Upper airway	***Symptoms/signs***	***Parenchymal***	***Symptoms/signs***	***Others***	***Symptoms/signs***
Cysts/masses: Vallecular cyst, cystic hygroma	*Vallecular cyst:* Stridor, suprasternal retractions, repeated extubation failures	Meconium aspiration syndrome	History of born through meconium stained amniotic fluid with patchy infiltrates in chest radiography	*Others:* Tracheoeso-phageal fistula (TEF), bilateral pulmonary hypoplasia, and anemia	High index of suspicion for TEF as many are undetected in antenatal scan Pulmonary hypoplasia is very common when rupture of membranes and oligohydramnios occur at 18–22 weeks of gestation
Laryngo-malacia	Presents beyond 2 weeks of age with stridor and symptoms of GERD				
Short neck syndromes causing glossoptosis	Dysmorphology				
Laryngeal stenosis or atresia	Stridor and difficult intubation				

(CNS: central nervous system; GERD: gastroesophageal reflux disease; IVH: intraventricular hemorrhage; PPHN: persistent pulmonary hypertension of the newborn; RD: respiratory distress; RDS: respiratory distress syndrome)

- ☑ Diagnosis of RDS is predominantly based on history and clinical symptoms. On clinical examination, neonates with RDS have tachypnea, grunting, nasal flaring, subcostal/intercostal retractions, cyanosis, and on auscultation, there might be decreased bilateral air entry.
- ☑ These symptoms progressively worsen in the first 24–48 hours, if inadequately treated or untreated. RDS can be associated with air leak syndromes and secondary persistent pulmonary hypertension of the newborn (PPHN).
- ☑ Though chest radiography is not mandatory to diagnose RDS or to decide on surfactant replacement therapy, it might be useful in relatively mature neonates of late preterm and term gestation in whom RDS is known to occur.

Antenatal Steroids and Delivery Room Continuous Positive Airway Pressure

- ☑ The most important aspect of management of RDS starts in the antepartum period with the administration of corticosteroids to mothers with threatened preterm labor.
- ☑ Dexamethasone is preferred over betamethasone in India as the evidence base for efficacy of betamethasone stems from a mixture of short- and long-acting salt, which is not available in India. For threatened labor between $24^{0/7}$ and $33^{6/7}$ weeks, dexamethasone at a dose of 6 mg intramuscular every 12 hourly for four doses is recommended.
- ☑ The efficacy of antenatal corticosteroids (ACS) is maximal when the delivery happens between 24 hours and 7 days after the last dose. While one rescue dose is advised in mothers who are at $<34^{0/7}$ weeks, had received one course of ACS 7–14 days prior and are at risk of delivering in the next 7 days, routine or scheduled repeat doses are not recommended.
- ☑ Though the use of ACS has been shown to decrease the risk of mortality or neurodevelopmental impairment at $23^{0/7}$–$23^{6/7}$ weeks, discussion with the parents is advised prior to its administration.
- ☑ There is no consensus for ACS use in late preterm pregnancies due to risk of neonatal hypoglycemia, and is not recommended for pregnancies between $37^{0/7}$ and $38^{6/7}$ weeks due to concerns regarding long-term neurodevelopmental disabilities at school age.

A recent systematic review had indicated that delivery room continuous positive airway pressure (CPAP) is associated with a decreased risk of invasive mechanical ventilation (IMV) as well as surfactant requirement in preterm neonates with RDS treated in low- and middle-income countries.

Noninvasive Respiratory Support and Invasive Mechanical Ventilation

- ☑ A recent network meta-analysis has shown that synchronized or non-synchronized noninvasive positive pressure ventilation (NIPPV) is superior to other noninvasive respiratory support (NRS) modalities when used as a primary mode of support in RDS.
- ☑ Among the other NRS, biphasic positive airway pressure (BiPAP) was found to be more efficacious than CPAP, which in turn was better than heated humidified high-flow nasal cannula (HHHFNC) in decreasing the risk of IMV.
- ☑ In case of NRS failure [fraction of inspired oxygen (FiO_2) > 0.60 to maintain an oxygen saturation of >90% or respiratory acidosis with a pH of <7.2 or repeated episodes of apnea], IMV may be initiated. IMV which is patient triggered with volume guarantee (4–6 mL/kg) has been shown to decrease many prematurity-associated complications.
- ☑ In case of IMV failure (requirement of peak inspiratory pressure of >25–28 cmH_2O and/or FiO_2 > 0.60 to maintain an oxygen saturation of >90% or presence of respiratory acidosis with a pH of <7.2), high-frequency ventilation is indicated.

Surfactant Replacement Therapy

- ☑ All babies with RDS who require FiO_2 > 0.30 to maintain an oxygen saturation of >90% with a CPAP level of ≥6 cmH_2O should be administered surfactant early, preferably within 2 hours of birth.
- ☑ Though there is no role for prophylactic surfactant administration, delivery room surfactant may be given if the neonate requires intubation in the delivery room for severe RD.
- ☑ Poractant alpha at an initial dose of 200 mg/kg is better than 100 mg/kg of beractant. Surfactant can be administered by either intubation-surfactant-extubate (INSURE) technique or less invasive surfactant administration (LISA).
- ☑ A recent large multicentric trial and multiple other systematic reviews have shown that LISA might decrease the risk of respiratory morbidities when compared to INSURE. A guide to choose the appropriate modality of surfactant administration is given in **Table 2**.

TABLE 2: Modality of surfactant administration—LISA versus INSURE versus surfactant followed by ongoing invasive mechanical ventilation.

LISA (all must apply)	***INSURE (all must apply)***	***Intubation and ongoing ventilation (any apply)***
Infants > 26 weeks (consultant discretion for infants < 26 weeks)	Infants < 26 weeks Infants > 26 weeks where ongoing ventilation may be required	–
Respiratory support only required at delivery	Respiratory support only required at delivery	Cardiac compressions or drugs required at resuscitation
Regular spontaneous respiration after caffeine administration	Regular spontaneous respiration after caffeine administration and following INSURE procedure	Poor respiratory efforts
FiO_2 > 0.3 but <0.6 at start of procedure and FiO_2 ≤ 0.3 at end of procedure	FiO_2 > 0.3 but <0.6 at start of procedure and ≤0.3 at end of procedure	FiO_2 ≥ 0.6 at start of procedure and/or >0.3 at end of procedure
Inotropic support < 10 µg/kg/min dopamine	Inotropic support < 10 µg/kg/min dopamine	Inotropic support ≥ 10 µg/kg/min dopamine
No other organ failure	No other organ failure	One or more organ failure

(FiO_2: fraction of inspired oxygen; INSURE: intubation-surfactant-extubate; LISA: less invasive surfactant administration)

Management

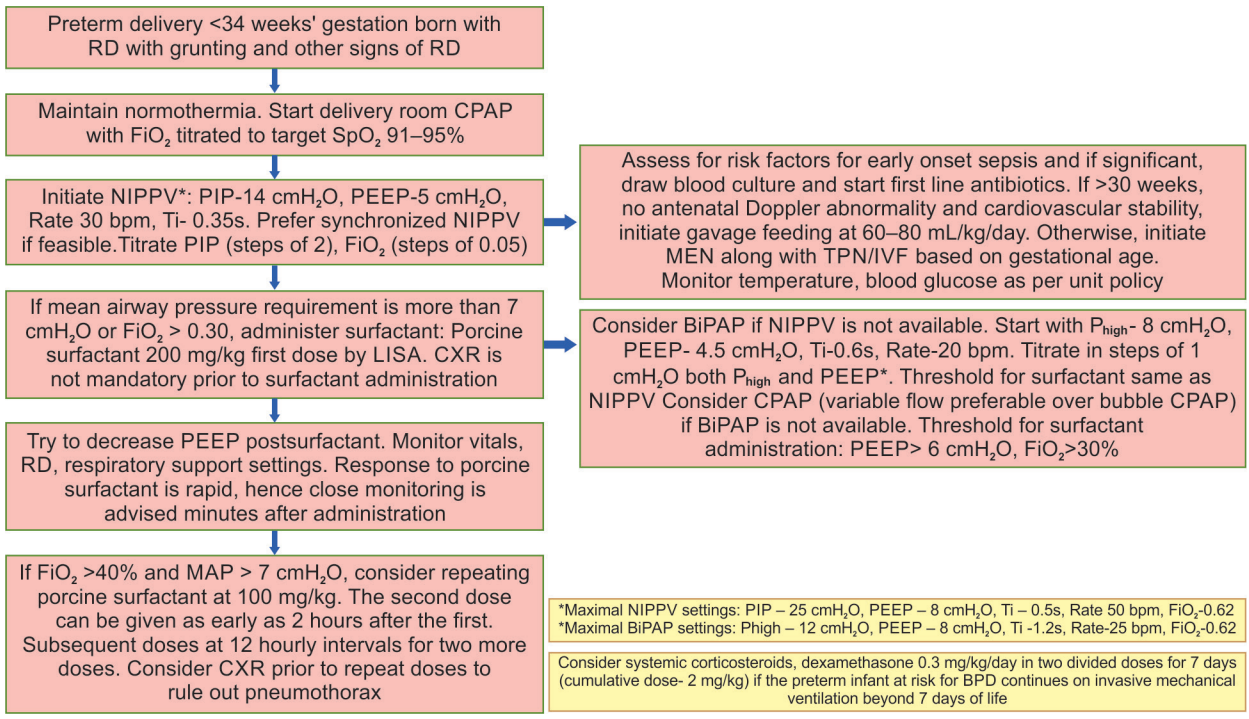
Supportive Care

- ☑ Maintaining normothermia immediately after birth and during neonatal intensive care unit (NICU) care is of vital importance.
- ☑ Use of plastic bag or wrap along with conventional care under a radiant warmer in the delivery room has been shown to improve temperature profile in very preterm neonates.
- ☑ Additional warming adjuncts such as a thermal mattress may also be utilized if the aforementioned modalities alone are not sufficient to maintain normothermia.
- ☑ Prophylactic caffeine therapy has been shown to be beneficial in decreasing the risk of mortality or bronchopulmonary dysplasia (BPD) and in improving long-term neurodevelopmental outcomes in preterm neonates of <30 weeks and/or <1,250 g. Recent literature has revealed that a higher dose of caffeine (40 mg/kg loading followed by 20 mg/kg/day maintenance) is more efficacious.
- ☑ Other routine management including maintaining euglycemia, adequate hematocrit level, serum calcium, and nutrition (enteral with or without parenteral) is also important.

A general approach to management of a neonate with RDS is shown in **Flowchart 1**. A graphical illustration of management of RDS in preterm neonates is depicted in **Figure 1**.

Management

Flowchart 1: Approach to a preterm neonate with RDS.



(BiPAP: biphasic positive airway pressure; CPAP: continuous positive airway pressure; CXR: chest X-ray; FiO_2: fraction of inspired oxygen; IVF: intravenous fluid; LISA: less invasive surfactant administration; MAP: mean airway pressure; tNIPPV: noninvasive positive pressure ventilation; PEEP: positive end-expiratory pressure; PIP: peak inspiratory pressure; RD: respiratory distress; RDS: respiratory distress syndrome; SpO_2: saturation of peripheral oxygen; TPN: total parenteral nutrition)

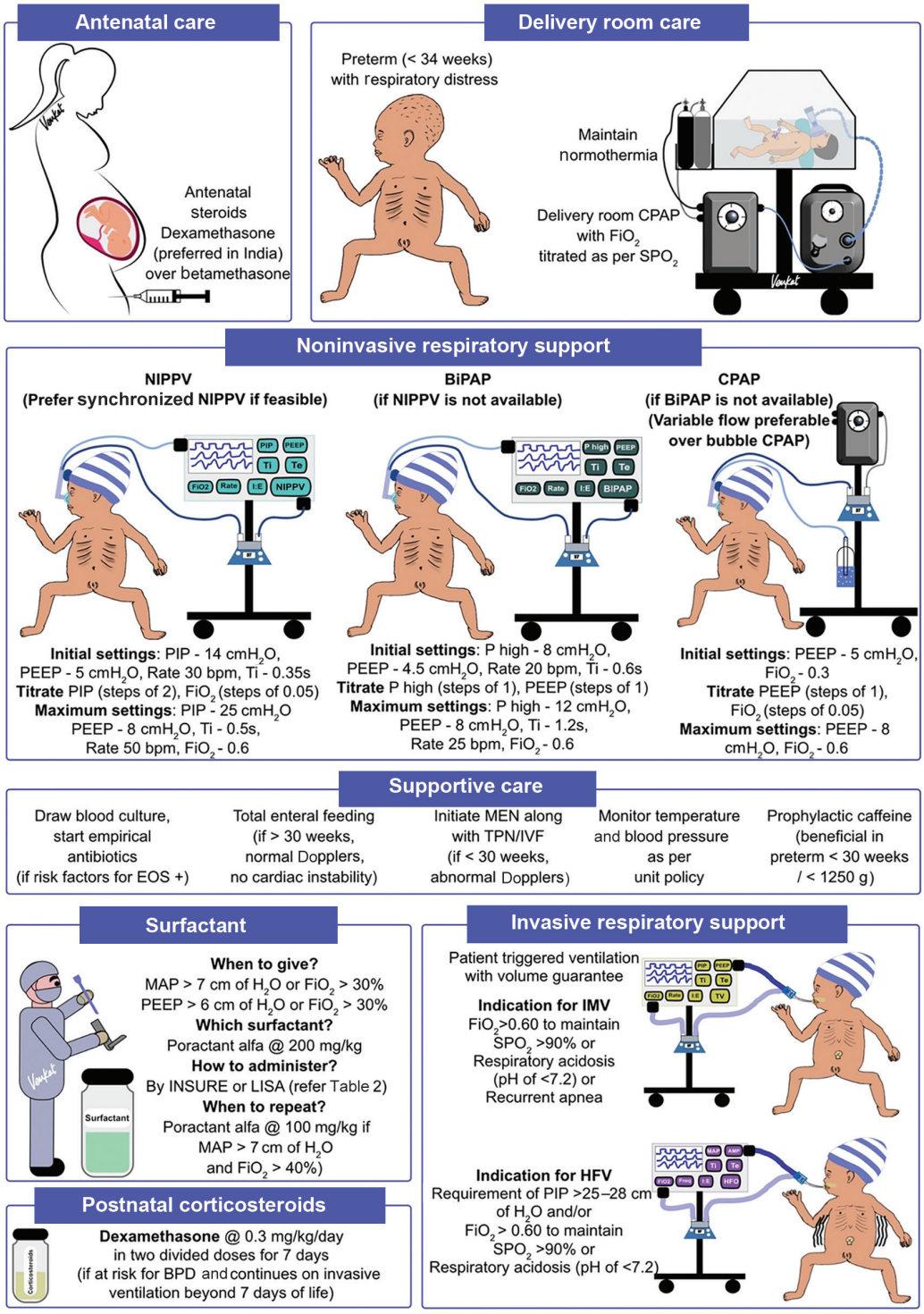


Fig. 1: Management of RDS in preterm neonates.

Courtesy: Dr Venkat Reddy Kallem, Paramitha Hospital, Hyderabad.

(BiPAP: biphasic positive airway pressure; CPAP: continuous positive airway pressure; FiO_2: fraction of inspired oxygen; HFV: high-frequency ventilation; INSURE: intubation-surfactant-extubate; IMV: invasive mechanical ventilation; IVF: intravenous fluid; LISA: less invasive surfactant administration; MAP: mean airway pressure; NIPPV: noninvasive positive pressure ventilation; PEEP: positive end-expiratory pressure; PIP: peak inspiratory pressure; RDS: respiratory distress syndrome; SpO_2: saturation of peripheral oxygen; TPN: total parenteral nutrition)

Further Reading

- ☑ Aldana-Aguirre JC, Pinto M, Featherstone RM, Kumar M. Less invasive surfactant administration versus intubation for surfactant delivery in preterm infants with respiratory distress syndrome: a systematic review and meta-analysis. Arch Dis Child Fetal Neonatal Ed. 2017;102(1):F17-23.
- ☑ Dargaville PA, Kamlin COF, Orsini F, Wang X, De Paoli AG, Kanmaz Kutman HG, et al. Effect of minimally invasive surfactant therapy vs sham treatment on death or bronchopulmonary dysplasia in preterm infants with respiratory distress syndrome: the OPTIMIST-A randomized clinical trial. JAMA. 2021;326(24):2478-87.
- ☑ Ramaswamy VV, Abiramalatha T, Bandyopadhyay T, Shaik NB, Pullattayil AK, Cavallin F, et al. Delivery room CPAP in improving outcomes of preterm neonates in low-and middle-income countries: a systematic review and network meta-analysis. Resuscitation. 2022;170:250-63.
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- ☑ Sweet DG, Carnielli V, Greisen G, Hallman M, Ozek E, Te Pas A, et al. European Consensus Guidelines on the management of respiratory distress syndrome—2019 Update. Neonatology. 2019;115(4): 432-50.

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Multisystem Inflammatory Syndrome in Children

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